

RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM - 602 105



**RAJALAKSHMI
ENGINEERING
COLLEGE**

SOFTWARE CONSTRUCTION
(CS 23432)

Laboratory Observation Note Book

Name : P.A.V.ITHRA...S.....

Year / Branch / Section : ... II. / B.Tech IT.....

Register No. : ... 241001162.....

Semester : ... IV.....

Academic Year : ... 2025 - 2026.....



**RAJALAKSHMI
ENGINEERING
COLLEGE**

**RAJALAKSHMI ENGINEERING COLLEGE (AUTONOMOUS)
RAJALAKSHMI NAGAR, THANDALAM – 602 105**

BONAFIDE CERTIFICATE

NAME PAVITHRA S REGISTER NO. 241001162

ACADEMIC YEAR 2025-26 **SEMESTER- IV** **BRANCH:** B. Tech. IT.

This Certification is the Bonafide record of work done by the above student

in the SOFTWARE CONSTRUCTION (CS23432) Laboratory

during the year 2025- 2026.

Signature of Faculty -in – Charge

Submitted for the Practical Examination held on 12.05.2026

Internal Examiner

External Examiner

INDEX

Name : PAVITHRA S Branch : IT Sec : _____ Roll No : 241001162

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AZURE DEVOPS ENVIRONMENTS SETUP

AIM:-

To set up and access the Azure DevOps environment by creating an organisation through the Azure portal.

INSTALLATION:-

- ① Open your web browser and go to the Azure site
<https://azure.microsoft.com/en-us/get-started/azure-portal>
Sign in using your Microsoft account credentials. ?
If you don't have a Microsoft account, you can create one here:
<https://azure//signup.live.com/2?ls=1>
Get started with Azure sign in
- ② Azure home page
Azure services - Azure DevOps organization.
- ③ Open DevOps environment in the Azure platform by typing Azure DevOps organisation in the search bar.
- ④ Click on the My Azure DevOps organisation link and create an organisation Home Page.
Azure DevOps.
My Azure DevOps Organizations.

RESULT:-

Successfully accessed the Azure DevOps environment and created a new organisation through the Azure Portal

AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

AIM :-

To setup an Azure Devops project for efficient collaboration and Agile work management.

PROCEDURE:-

- ① Create an Azure Account
Azure Devops
pavithra.s.2024.it@vjalandra.edu.in
- ② Create the first project in your organisation
dev.azure.com / Student Data Management System
- a) After the organisation is setup, you'll need to create your first project. This is where you'll begin to manage code, pipelines, work items and more.
- b) On the organisation's Home page, click on the New project button
- c) Enter the project name, description and visibility.
Name: Choose a name for the project
Description: Optionally, add a description to provide more context about the project.
Visibility: Choose whether you want the project to be private or public.
- d) Once you've filled out the details, click create to set up your first project.

Create new project

Project name: Student Data Management System

Visibility: Public

Version Control: Git

Work item process:

- ③ Once logged in, ensure you are in the correct organisation. If you're part of Multiple Organisation you can switch between them from the top left corner (next to your user profile). Click on the organisation name, and you should be taken to the Azure DevOps Organisation Home Page.

~~Pa~~
Projects:-

- ④ Project dashboard

About this project:

The Student Data Management System (edutrack) is a web-based application used to manage student records efficiently. It allows instructors to add, update, delete and view student details. The system supports bulk data import using CSV files with validation. It also provides search functionality for quick access to student information.

- ⑤ To Manage User Stories:-

User stories define features like adding students, uploading CSV files, and searching records. In Azure DevOps, these are managed using Boards and work items. Each story represents a function of the system and can be assigned and tracked. This helps in organizing and completing tasks efficiently.

Boards → Work items → + new work items → user story

RESULT:-

Successfully created on Azure DevOps project with user story management and Agile workflow setup.

Setting up Epics, Features and User Stories for Project Planning

AIM:-

To learn about how to create epics, user story, features, back logs for your assigned project.

PROCEDURE:-

Create epic, features, user stories, task.

① Fill in epic

 Epic 1

Student Data Management and Batch Processing System.

② Fill in Features:

 User Registration and Login

 Student Record Management (CRUD)

 CSV File Upload for Bulk Import

 Student Search Functionality

 Data Validation and Error Handling

③ Fill in User Story Details:-

 Student Record Management:

As an instructor, I want to add, update, and delete student details so that I can manage student records efficiently.

 CSV Upload System:-

As an instructor, I want to upload CSV files so that I can import multiple student records at once.

 Student Search System:-

As an instructor, I want to search student details so that I can quickly find required information.

RESULT:-

Thus, the creation of epics, features, user stories, and tasks has been completed successfully.

A red checkmark with a stylized signature or flourish at the end of the stroke.

SPRINT PLANNINGAIM:-

To assign user stories to specific sprints for the Student Data Management System

Sprint PlanningSprint 1: Taskboard

1. user Registration and Login As

Assigned to :

Iteration 1: Taskboard

2. Add and view student Records

Assigned to :

Iteration 2: Taskboard

3. Update and Delete student Records

Assigned to :

Iteration 3: Taskboard

4. CSV File upload for Bulk Import

Assigned to :

5. Student Search Functionality

Assigned to :

RESULT:

The sprints are created for the Student Data Management System.

POKER ESTIMATIONAIM:-

To create poker estimation for the user stories Student data Management System.

POKER ESTIMATION:-

story points measure:

- ★ Effort
- ★ Complexity
- ★ Time

This uses Fibonacci scale: 1, 2, 3, 5, 8, 13, ...

Story points for Student Data Management System.

Story w	Story points	Reasons
User Registration	3	Form handling + data storage
Student Record Management (Add/View)	5	CRUD operations + database interaction
Update and Delete Student Records	5	Data modification + validation
CSV File upload	8	File handling + validation + bulk processing
Student Search Functionality	3	Query + filtering Logic

Total = 24 story points

~~Task~~ Boards → work items → user story → Story points
(1, 2, 3, 5, 8)

→ same repeat for all user stories

RESULT:-

The estimation / story points are created for the project using Poker Estimation.

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

AIM:-

To design a class diagram and sequence diagram for the Student Data Management System.

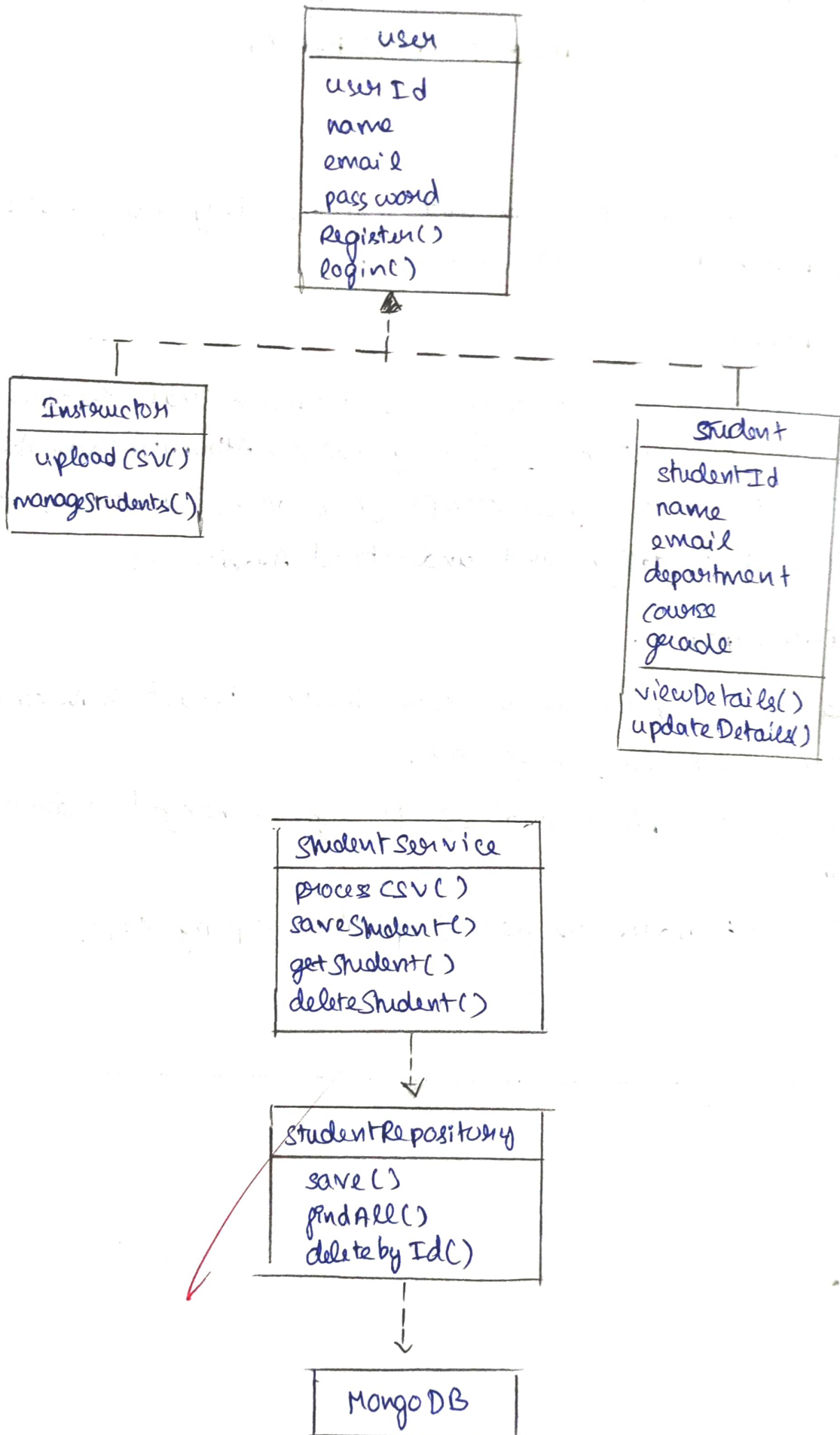
CLASS DIAGRAM:-

- * A class diagram is a type of UML diagram that represents the structure of a system by showing classes - Attributes (variables), Methods (functions), Relationships (Association, Inheritance, etc---)
- * It is used to design and understand Architecture

SEQUENCE DIAGRAM:-

- * A sequence diagram shows how objects interact with each other in a particular sequence of time.
- * It represents object involved, Message exchanged, order of execution.
- * It is used to model the flow of operation step by step.

CLASS DIAGRAM :-



SEQUENCE DIAGRAM:-

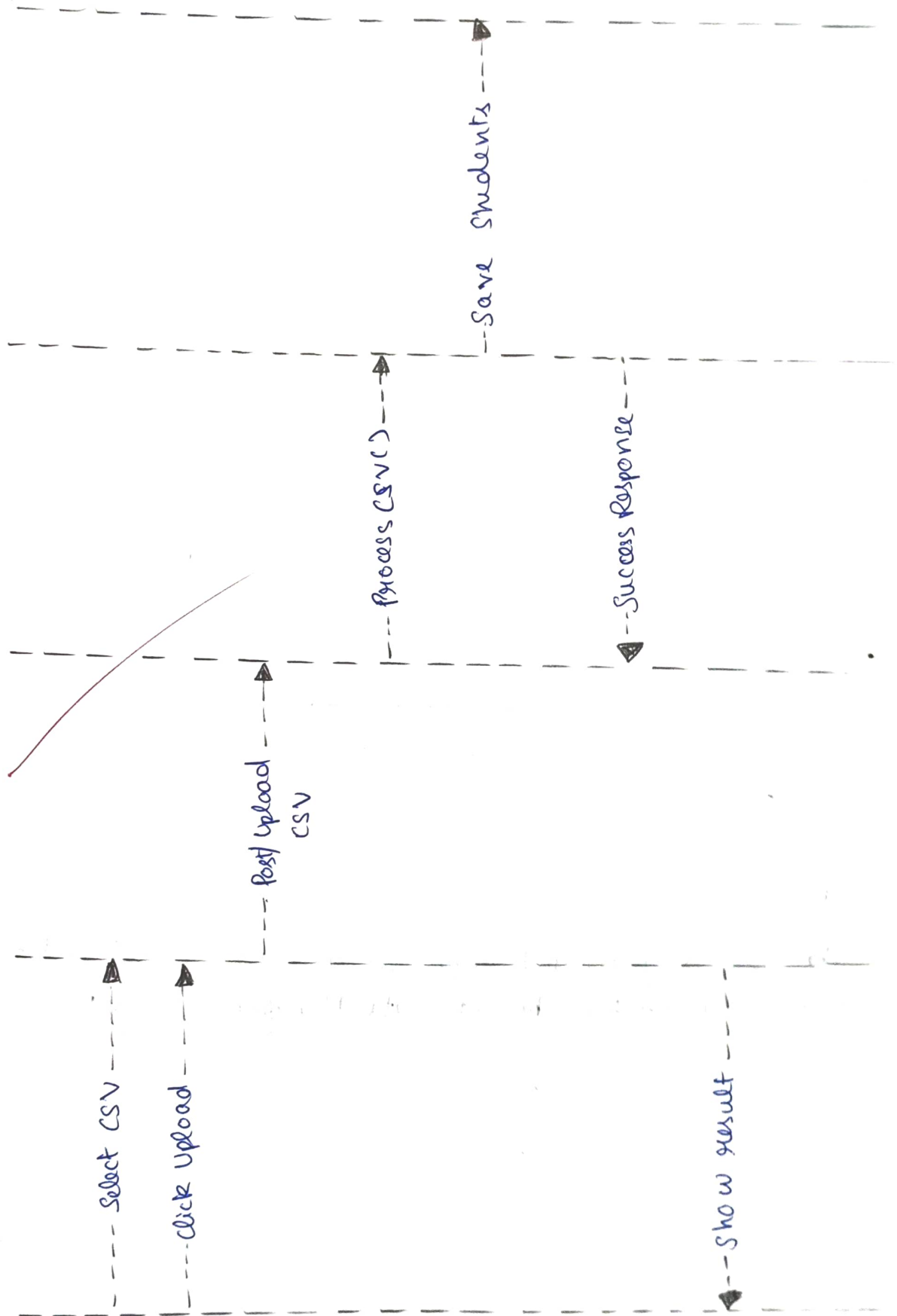
INSTRUCTOR

FRONTEND (UI)

BACKEND (API)

SERVICE

DATABASE



RESULT:-

The class diagram and sequence diagram is designed & successfully for the Student Data Management System.



DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

AIM:-

To design an architectural diagram and ER diagrams for the given project (Student Data Management System).

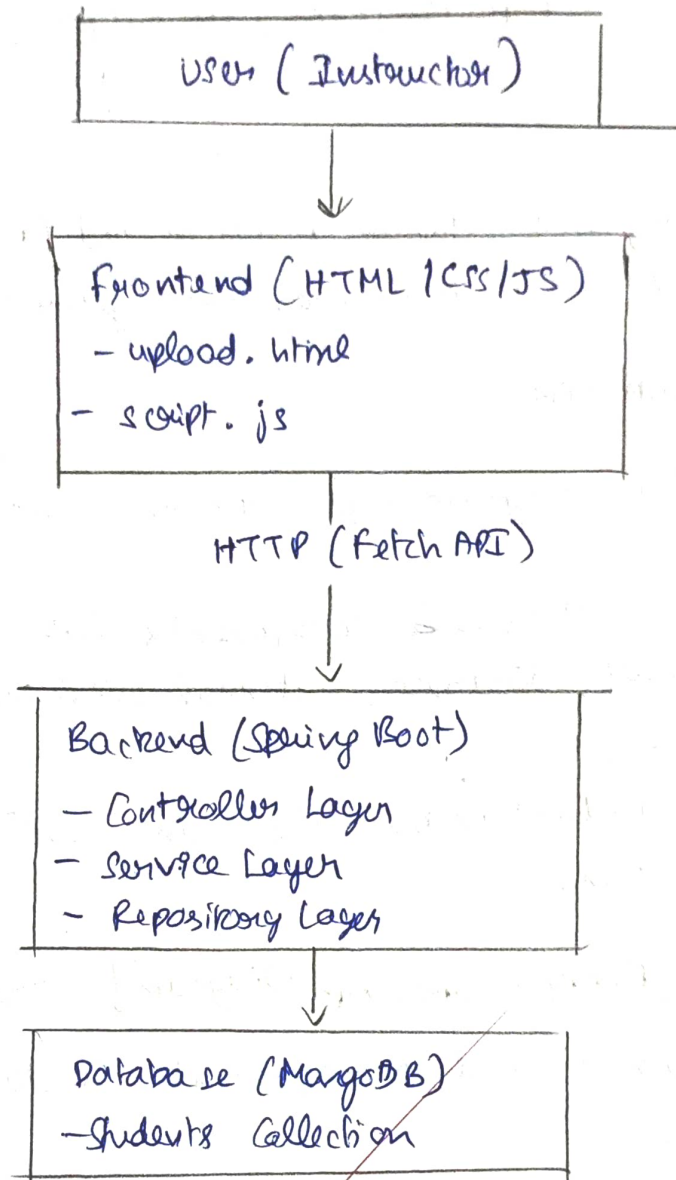
ARCHITECTURAL DIAGRAM:-

- ③ ★ An architectural diagram represents the overall structure of a system.
- ★ It shows how different components like user interface, Application/server, database, external services interact with each other.
- ★ It helps in understanding the high-level design and data flow of the system.

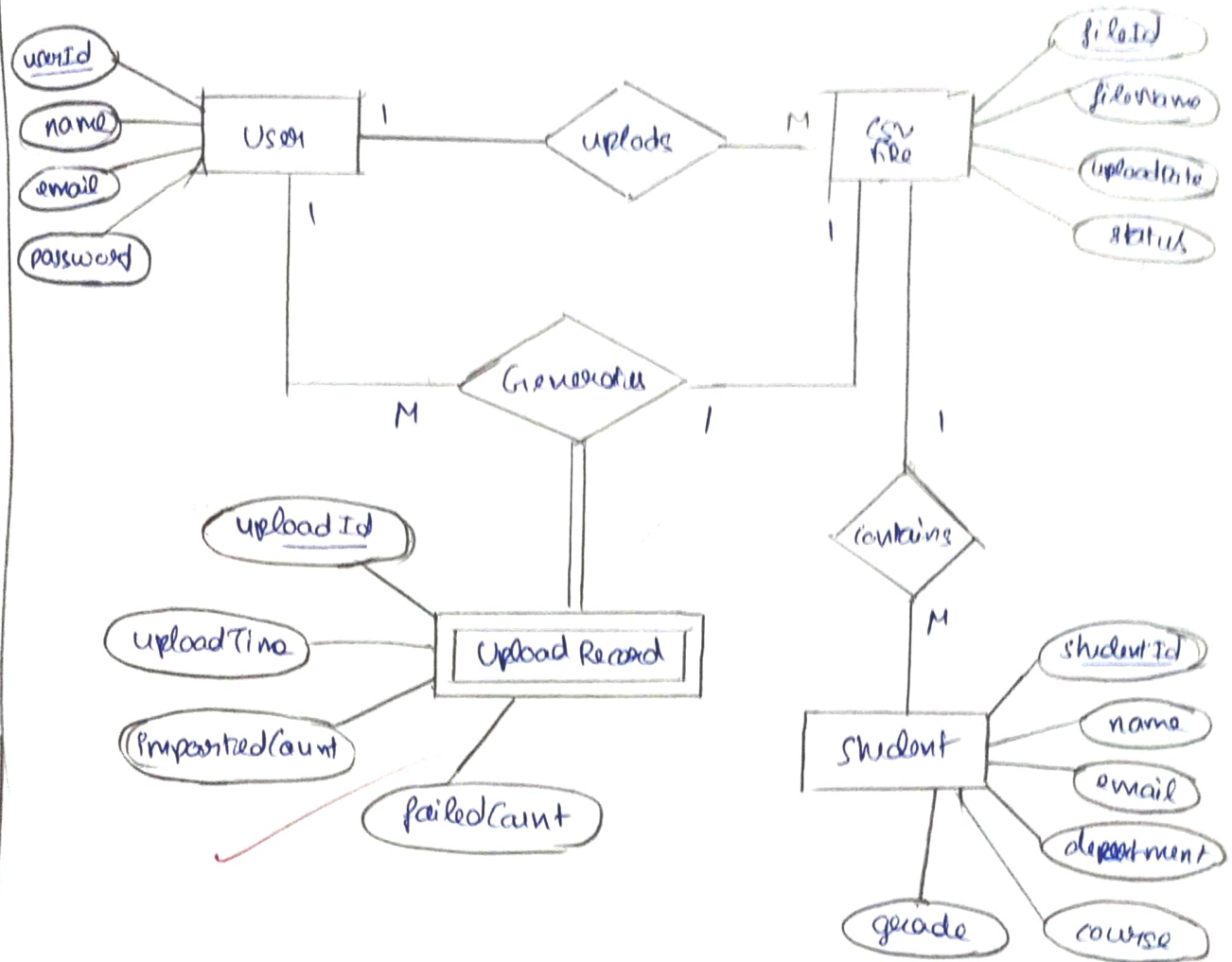
ER DIAGRAM:- [Entity Relationship Diagram]

- ★ An ER diagram is used to represent the database structure of a structure.
- ★ It shows entities (tables), attributes (fields), relationships between entities.
- ★ It helps in designing and organising the database schema.

ARCHITECTURE DIAGRAM :-



ER DIAGRAM:



RESULT:-

Thus, the Architectural diagram and ER Diagram are designed successfully for Student Data Management System.

TESTING - TEST PLAN AND TEST CASES

AIM:

Test plans and test cases for Student Data Management System and write at least five user stories showcasing happy path and error scenarios.

Test planning and Test Case Design Procedure :-

① Understand core features of the application :-

- User registration and login
- Admin dashboard access
- Student record management
- CSV file upload for bulk student import
- View and manage uploaded student data.

② Define user interactions

- Simulate user actions such as registration, login, adding students, uploading CSV files, and viewing records.
- Include incorrect login credentials, invalid CSV upload, missing student details, and unauthorized access.

③ Design happy path and error path test cases.

- Focus on validating features under normal conditions.
Eg:- Admin uploads valid CSV and student data is stored successfully.

④ Design error path test cases.

- Simulate negative scenarios to test robustness.
Eg:- Login fails with incorrect password, invalid file type upload, empty CSV file, duplicate student ID.

⑤ Breakdown steps and expected results.

- Each test case includes clear actions and expected outcomes.
- Ensures clarity for both manual and automated testing.

⑥ Use clear naming and IDs.

TC01 - Successful Registration

TC02 - Invalid Login

TC03 - CSV Upload Success

TC04 - Invalid CSV upload.

⑦ Separate Test Suites:-

Authentication

Student Management

CSV Upload

Dashboard

⑧ Prioritize and review:-

Critical actions like login, registration, and CSV upload are marked as high priority

Test Suite:-

TS01 - User Authentication

→ TC01 - Successful Registration

Action:

- Open register page
- Enter valid username, email, password
- Click "Register"

Expected Result:

Account created successfully.

User redirected to login page.

Type: Happy Path.

→ TC02 - Registration with existing Email.

Action:

- Open register page
- Enter already registered email
- Click "Register"

Expected Result:-

Error message: "Email already exists"

Type: Error path.

→ TC03 - Successful Login

Action:

- Open Login Page
- Enter valid email and password
- Click "Login"

Expected Result:

User redirected to dashboard

Type: Happy Path.

→ TC04 - Login with Invalid Password

Action:

- Open Login page
- Enter valid email + wrong password
- Click "Login"

Expected Result:-

Error message: "Invalid credentials"

Type: Error Path

Test Suite TS02 - Student Management

→ TC05 - Add Student Successfully

Action:

- Navigate to Add Student page
- Enter valid student details
- Click "Save"

Expected Result:

Student added successfully

Student appears in student list.

Type: Happy Path.

→ TC06 - Add Student with Missing Fields

Action:

- Leave mandatory fields blank.
- Click "Save"

Expected Result:

Validation message shown

Record not saved

Type: Error Path

Test Suite TS03 - CSV Upload

→ TC07 - Upload Valid CSV file

Action:

- Navigate to Upload CSV
- Select valid CSV file
- Click "Upload"

Expected Result:

File uploaded successfully,

Student records imported into database.

Type: Happy Path.

→ TC08 - Upload Invalid File Format

Action:

- Select non - CSV File
- Click "Upload"

Expected Result:-

Error message: "Only CSV files allowed"

Type: Error Path

→ TC09 - Upload Empty CSV

Action:

- Select empty CSV file
- Click "Upload"

Expected Result:

Error message displayed

No records imported

Type: Error Path

Test Suite TS04 - Dashboard :-

→ TC10 - View Dashboard Successfully

Action:-

- Login successfully
- Open dashboard

Expected Result:-

Dashboard loads with student statistics

Type: Happy Path

→ TC11 - Unauthorized Dashboard Access

Action:

- Open dashboard without login

Expected Result:

Redirect to login page

Type: ~~S~~ Broken Path

Test Suite TS05 - Logout.

→ TC12 - Successful Logout

Action:

- Click logout button

Expected Result:

Session cleared

Redirected to login page

Type: Happy Path

USER STORIES:

- ★ As an ~~Admin~~ Admin, I want to register securely so that I can access the system.
- ★ As an Admin, I want to log in ~~securely~~ so that I can manage student records.
- ★ As an Admin, I ~~should~~ be able to add student details manually when required.
- ★ As an admin, I should be able to upload CSV files to bulk import student records.
- ★ As an admin, I need ~~secure~~ logout to protect system data.

LOAD TESTING AND CI/CD PIPELINE

AIM:-

To perform load testing on the EduTrack Student Data Management System web application to evaluate its performance under multiple users, and to implement a CI/CD pipeline for automating the build, testing, and deployment process of the application.

LOAD TESTING:-

- * The Load Testing was performed on the EduTrack / Student Data Management System developed using Spring Boot, MongoDB, HTML, CSS and JavaScript.
- * Multiple virtual users were simulated to access the application simultaneously including:
 - User registration
 - User login
 - Student Record Management
 - CSV upload
 - Dashboard Access.

PIPELINE SETUP:-

- * A CI/CD pipeline was configured using Azure DevOps by connecting the Github repository containing the EduTrack project
- * The pipeline automates:
 - Code checkout from github
 - Java / Maven environment setup
 - Dependency installation
 - Build process
 - Application execution / testing

YAML:

trigger:

- main

Pool:

vmImage: 'ubuntu - latest'

Steps:

- Checkout: self

- task: Java Tool Installer @ 0

Inputs:

versionSpec: '17'

jdk Architecture Option: 'x64'

jdk Source Option: 'pre installed'

displayName: 'set up Java'

- script: |

~~main~~ clean install
mvn

displayName: 'Install dependencies and build project'

- script: |

mvn spring-boot:run

displayName: 'Run Spring Boot Application'

RESULT:-

The Student Data Management System successfully handled multiple user ~~xxx~~ request during load testing with stable performance, the CI/CD pipeline was successfully implemented, automating project build, dependency installation, testing, and execution ~~ensuring~~ continuous integration and smoother deployment.

GITHUB PROJECT STRUCTURE AND NAMING CONVENTIONS

AIM:-

To organize the Student Data Management System project using a proper folder structure and naming conventions in Github, making the project easy to understand, maintain and extend.

GITHUB:-

The project was ~~at~~ uploaded to a Github repository for version control and collaboration. One repository includes all source code, frontend files, backend files, configuration files and doc, allowing easy access, updates, and project management.

PROCEDURE:-

- ① Created the Student Data Management System project using Spring Boot, MongoDB, HTML, CSS, and JavaScript.
- ② Organised files into proper folders such as:
 - * frontend / for UI pages
 - * ~~src~~src/main/java/com/edutrack/ for backend source code.
 - * controller / for request handling
 - * service / for business logic
 - * repository / for db operations
 - * model / for data models
 - * resources / for configuration files
- ③ Followed proper naming conventions for
 - * Java classes (student.java, user.java)
 - * Controllers (studentController.java)
 - * Services (studentService.java)
 - * Repository interfaces.
 - * HTML Pages (register.html, login.html, upload.html)

- 1) Created a GitHub repository for EduTrack Project.
- 2) Uploaded all project files and maintained structured commits for version tracking.
- 3) Verified the project structure, readability and maintainability in the repository.

RESULT:-

The EduTrack Student Data Management System project ~~was~~ was successfully organised using a clear folder structure and proper naming conventions. The GitHub repository clearly displays the project architecture, making it easy for developers and users to understand, maintain, and navigate the codebase efficiently.

